31. Prove that there is no positive integer n such that $n^2+n^3=100$.

Solution: We shall solve this problem using a proof by cases:

Case 1: $n=1$ ∴ $n^2+n^3=1^2+1^3=1+1=2<100$

Case 2: $n=2$ ∴ $n^2+n^3=2^2+2^3=4+8=12<100$

Case 3: $n=3$ ∴ $n^2+n^3=3^2+3^3=9+27=36<100$

Case 4: $n=4$ ∴ $n^2+n^3=4^2+4^3=16+64=80<100$

Case 5: $n\geq 5$. $5^3=125>100$. ∴ It's easy to verify that for all $n\geq 5$, $n^3>100$. ∵ For any arbitrary integer n, $n^2\geq 0$.

∴ $n^3+n^2>100$ for all $n\geq 5$.

∴ There is no positive integer n such that $n^2+n^3=100$ □

32. Prove that there are no solutions in integers x and y to the equation $2x^2+5y^2=14$.

Solution: WLOG, we can assume the integers are non-negative, ∵ for any arbitrary integer n, $n^2\geq 0$.

We shall give a proof by cases:

Case 1: $x=0, y=0$. ∴ $2x^2+5y^2=2\cdot(0)^2+5\cdot(0)^2=0+0=0<14$

Case 2: $x=0, y=1$ ∴ $2x^2+5y^2=2\cdot(0)^2+5\cdot(1)^2=0+5=5<14$

Case 3: $x=0, y\geq 2$ $2x^2+5y^2\geq 2\cdot(0)^2+5\cdot(2)^2=0+20=20>14$

Case 4: $x=1, y=0$. ∴ $2x^2+5y^2=2\cdot(1)^2+5\cdot(0)^2=2+0=2<14$

Case 5: $x=1, y=1$. ∴ $2x^2+5y^2=2\cdot(1)^2+5\cdot(1)^2=2+5=7<14$